The Sperm Parasite Paradox

Stochastic Stabilization in Gynogenetic Complexes

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The Sperm Parasite Paradox

The logic of collapse:

1. Each male–parasite mating produces *only* parasite offspring
2. Growing parasite population diverts male effort from host females
3. Fewer host matings ⇒ fewer males next generation
4. Fewer males ⇒ less sperm for everyone
5. Positive feedback ⇒ **extinction**

Yet: *Poecilia formosa* has coexisted with *P. mexicana* and *P. latipinna* for > **100,000 years** *(Avise 1991)*.

Key biology

P. formosa = **sperm parasite**

Requires host male sperm to trigger embryogenesis

Paternal genome **excluded** from all offspring

All-female species: 100% parasite daughters

Field Evidence: Behavioral Discrimination

Riesch, Schlupp & Plath (2008):
Sperm acquisition in natural populations of *P. latipinna* + *P. formosa*

Species	% with sperm
Host females	**74%**
Parasite females	**10%**

- Discrimination is **behavioral**, not physiological
- Males allocate sperm *actively*
- Discrimination persists even when males are exceedingly rare
- Fecundity of lab-reared *P. formosa* is normal when sperm is given

Mechanism

Upon each parasite encounter, host male accepts with probability $\gamma \in (0, 1]$

γ = **per-encounter acceptance probability**

Not a vague "discrimination factor"—a measurable behavioral parameter

Environmental context

Soto La Marina drainage: seasonal flooding, pool isolation/reconnection

High year-to-year variation in parasite proportion $\Rightarrow$ **stochastic forcing is real**

Prior Models: What Is Missing

Study	Contribution	Gap
Kiester (1981)	Discrimination necessary for persistence	Fixed γ; no 3-population structure
Schley (2004)	Ecological factors + Allee effects	No parasitism
Peck (1999)	Spatial fragmentation matters	No discrimination behavior
Kokko (2008)	Spatial dynamics + sperm limitation	γ not population-dependent
Heubel (2009)	Male efficiency + fixed discrimination	Fixed γ; no stochasticity

Common gaps in all prior models:

- No population-*dependent* discrimination function $\gamma(p)$
- No systematic comparison of competing stochastic formulations
- Discrimination *speed* never identified as the key control variable

Mechanistic Derivation: Two Steps

State: host females f, host males m, parasite females p

Step 1 — Per-encounter acceptance

Mass-action encounters; upon each parasite encounter, male accepts with probability γ:

host matings: $\alpha\, m\, f$

parasite matings: $\gamma\, \alpha\, m\, p$

Independent pools at this stage.

Step 2 — Finite mating capacity

Each accepted parasite mating *displaces* one host-female mating:

$$\alpha mf - \gamma\alpha mp = \alpha m(f - \gamma p)$$

Key result

$m(f - \gamma p)$ is a **theorem** of Steps 1 & 2, not an assumption.

Critics of the functional form must challenge Step 2 specifically: *Is male mating capacity finite?*

Riesch (2008): yes — sperm allocation is a zero-sum budget.

Simplifying assumption: Displacement coefficient $c = 1$ (future: calibrate from sperm-count data)

The Deterministic System

Dimensional system (tildes = dimensional quantities):

$$\frac{d\tilde{f}}{dt} = a\,\tilde{\beta}\,\tilde{L}\,\tilde{m}(\tilde{f} - \gamma\tilde{p}) - \tilde{\delta}\,\tilde{f}$$

$$\frac{d\tilde{m}}{dt} = (1-a)\,\tilde{\beta}\,\tilde{L}\,\tilde{m}(\tilde{f} - \gamma\tilde{p}) - \tilde{\delta}\,\tilde{m}$$

$$\frac{d\tilde{p}}{dt} = \gamma\,\tilde{\beta}\,\tilde{L}\,\tilde{m}\,\tilde{p} - \tilde{\delta}\,\tilde{p}$$

where $\tilde{L} = 1 - (\tilde{f} + \tilde{m} + \tilde{p})/\tilde{K}$

Non-dimensionalize: $u_i = \tilde{u}_i/\tilde{K}$, $\tau = \tilde{\delta}\,t$, $\beta = \tilde{\beta}\tilde{K}/\tilde{\delta}$

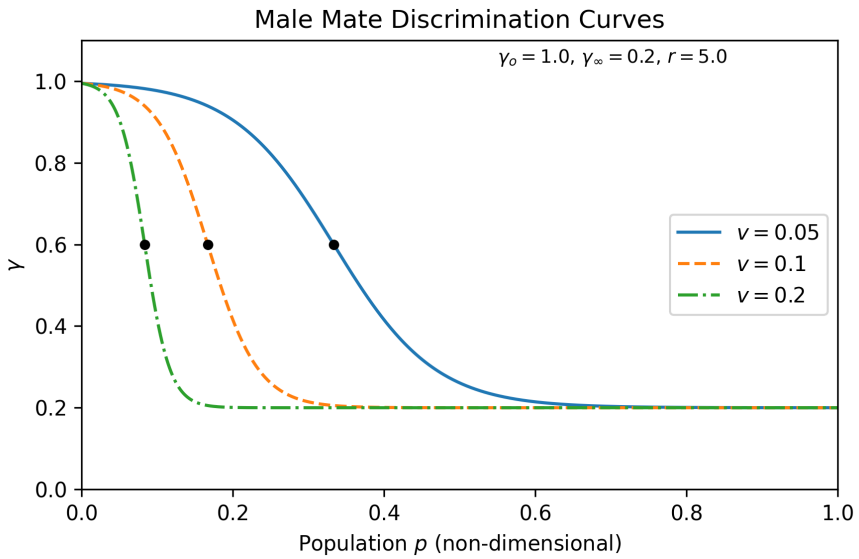
Parameter	Symbol	Value	Source
Birth rate (nondim)	β	300	$0.1 \times 300/0.1$
Female fraction	a	0.8	Balsano (1989)
Carrying capacity	$\tilde{K}$	300	field estimate
Discrim. (initial)	γ_o	1.0	no preference
Discrim. (asymp.)	γ_∞	0.2	Riesch (2008)

Population-Dependent Discrimination $\gamma(p)$

Sigmoid function:

$$\gamma(p) = \frac{\gamma_o - \gamma_\infty}{1 + e^{vp-r}} + \gamma_\infty$$

- $\gamma_o = 1$: no preference when parasites rare
- $\gamma_\infty = 0.2$: strong discrimination when parasites common
- $v > 0$: **speed of discrimination**
- $r = 5$: sigmoid shift



Three discrimination speeds. Black dots: inflection points.

v is the pivotal parameter:
slow $v \Rightarrow$ extinction
fast $v \Rightarrow$ coexistence

Deterministic Fate: Extinction Without Noise

Equilibrium condition:

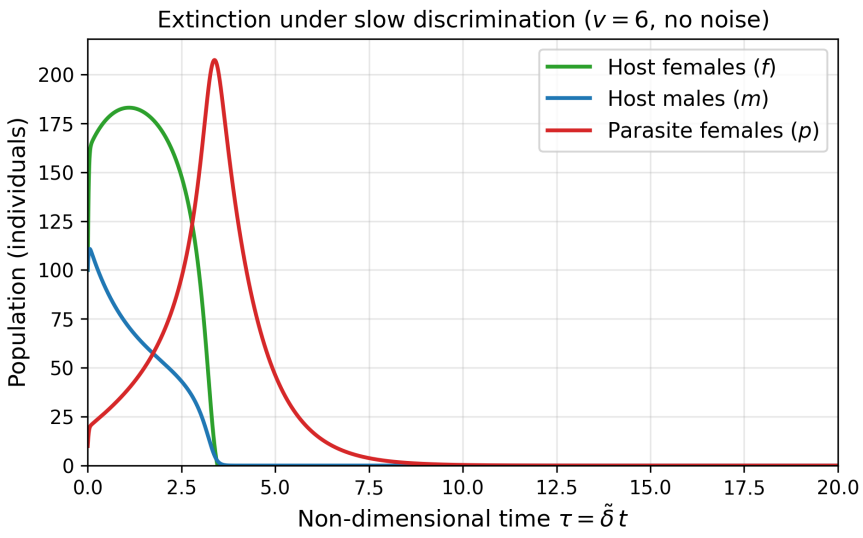
$$\gamma_{eq} = \frac{a\,f^*}{a\,p^* + f^*} \approx 0.74$$

Three regimes:

- $\gamma < \gamma_{eq}$: **extinction**
- $\gamma = \gamma_{eq}$: coexistence
- $\gamma > \gamma_{eq}$: parasites extinct, hosts recover

Slow discrimination ($v = 6$, nondim): parasite initially grows, draws down host males, starves host females, eventually collapses itself.

Without noise: no escape.
With noise: ?



All three populations $\to 0$. No noise, $v = 6$, $\tau = 20$ ($\approx$200 days).

Why Stochasticity?

Environmental evidence:

- Soto La Marina: seasonal flooding creates alternating isolation and connectivity
- During dry season: disconnected pools
- During rainy season: pools reconnect
- Field data (Balsano 1989): *wide* year-to-year variation in parasite proportion *at the same location*

Stochastic forcing is not a modeling convenience.
It is a biological fact in this system.

Three modeling frameworks

RODE: piecewise-constant parametric noise; smooth paths

Itô SDE: continuous multiplicative Wiener noise; left-endpoint integral

Stratonovich SDE: same noise structure; midpoint integral; Wong-Zakai limit of smooth environmental forcing

Each noise structure: **common** (shared W) or **independent** (per-population W_i)

Five Formulations — One Biological System

#	Formulation	Noise	Biological interpretation
1	RODE	Piecewise-constant	Discrete environmental pulses
2	Itô / common	Single $W(t)$	Shared habitat driver
3	Itô / independent	W_f, W_m, W_p	Independent fluctuations
4	Stratonovich / common	Single $W(t)$	Limit of smooth forcing
5	Stratonovich / indep.	W_f, W_m, W_p	Limit of smooth, independent

The central question of this talk

These are not five ways to write the same model.
They encode **different physical assumptions** and produce **different quantitative predictions** for extinction probability.

A researcher who picks one and reports it as "the stochastic model" is presenting **one answer among five**.

RODE: Random Ordinary Differential Equation

Noise enters as piecewise-constant perturbation to death rate:

$$\frac{df}{d\tau} = a\beta Lm(f - \gamma p) - (1 - \eta_f)f$$

$\eta_i(t) \sim \mathrm{Uniform}[0, \tilde{\eta}_{\max}]$, resampled each step.

Key property: between resampling events, system is a standard ODE $\Rightarrow$ classical RK45 applies.

Noise structure (from Balsano 1989 association data):

$$\eta_f = \eta_m = \tfrac{2}{3}\eta, \quad \eta_p = \eta$$

Parasite females bear full environmental load; host populations bear two-thirds.

RODE Boundary Dissipativity Lemma

On the population ceiling $f + m + p = 1$:

$$\nabla \cdot \mathbf{F}\big|_{f+m+p=1} \leq \eta_f + \eta_m + \eta_p - 3$$

System is **dissipative** at the ceiling whenever:

$$\eta_f + \eta_m + \eta_p < 3$$

i.e., noise amplitudes sum below the death rate threshold.

Proof: At $L = 0$, all birth terms vanish.
Liouville's theorem gives phase-space contraction directly.

Itô SDE: Multiplicative Geometric Noise

Common noise (single Wiener process W):

$$df = \mu_f(\mathbf{u})\,dt + \sigma_f f\,dW, \quad dm = \mu_m(\mathbf{u})\,dt + \sigma_m m\,dW, \quad dp = \mu_p(\mathbf{u})\,dt + \sigma_p p\,dW$$

Key analytical result — mean equations (Proposition 1):

Itô noise does NOT shift ensemble means

Under mean-field closure, $F = \mathbb{E}[f]$, $M = \mathbb{E}[m]$, $P = \mathbb{E}[p]$ satisfy *exactly the deterministic ODE*.

Proof: the stochastic integral $\int \sigma_f f\,dW$ is a martingale; $\mathbb{E}[\Delta W \mid \mathcal{F}_t] = 0$ eliminates the noise term.

Scalar Lyapunov exponent for parasite p-subsystem:

$$\lambda_{\text{Itô}} = \beta_{\text{eff}} - \delta - \tfrac{1}{2}\sigma_p^2$$

Extinction ($\lambda < 0$) requires $\sigma_p > \sqrt{2(\beta_{\text{eff}} - \delta)}$.
Increasing Itô noise promotes extinction of the parasite.

Stratonovich and Wong-Zakai: What Do They Mean?

The Itô vs. Stratonovich question

Both write $dX = \mu\, dt + \sigma X\, dW$.

They differ in **when** the integrand is evaluated inside $\int g(X)\, dW$:

- **Itô:** left endpoint of each interval
  $\Rightarrow$ integrand evaluated *before* the noise kick
- **Stratonovich:** midpoint
  $\Rightarrow$ integrand evaluated *during* the noise kick

For **additive** noise: identical.
For **multiplicative** noise ($g = \sigma X$):
the midpoint rule adds an extra drift term $+\frac{1}{2}\sigma^2 X$.

Same equation. Different calculus. Different physics.

The Wong-Zakai theorem in plain language

Question: Real environmental noise is never truly white. It is *colored* — it has a finite correlation time τ_c (e.g., a flood lasts days, not an instant).

What happens as $\tau_c \to 0$?

Replace white noise $W(t)$ with a smooth approximation $W^\epsilon(t)$ with correlation time ϵ, solve the ODE, then take $\epsilon \to 0$.

Wong & Zakai (1965): the limit is the **Stratonovich** SDE, not the Itô SDE.

Plain conclusion: if your noise is a physical process smoothed over a short time scale — seasonal flooding, temperature oscillations, pool connectivity — **Stratonovich is the correct limit, not a convention.**

For *P. formosa*: Soto La Marina flooding has correlation time $\sim$ weeks to months $\Rightarrow$ Stratonovich is physically motivated, Itô is a mathematical convenience.

Stratonovich SDE: The Wong-Zakai Limit

Stratonovich formulation (midpoint integral):

$$df = \mu_f \, dt + \sigma_f f \circ dW$$

Itô–Stratonovich conversion adds drift correction $+\frac{1}{2}\sigma_i^2 u_i$ per population.

Stratonovich noise DOES shift ensemble means (Proposition 2)

$$\frac{dF}{dt} = a\beta(1 - F - M - P)M(F - \gamma P) - \delta F + \frac{1}{2}\sigma_f^2 F$$

The noise-induced increment $+\frac{1}{2}\sigma_i^2 u_i$ acts as an **effective noise-induced birth rate**. Since $\sigma_p > \sigma_f = \sigma_m$ (parasite bears more noise), the parasite receives the **largest growth bonus**.

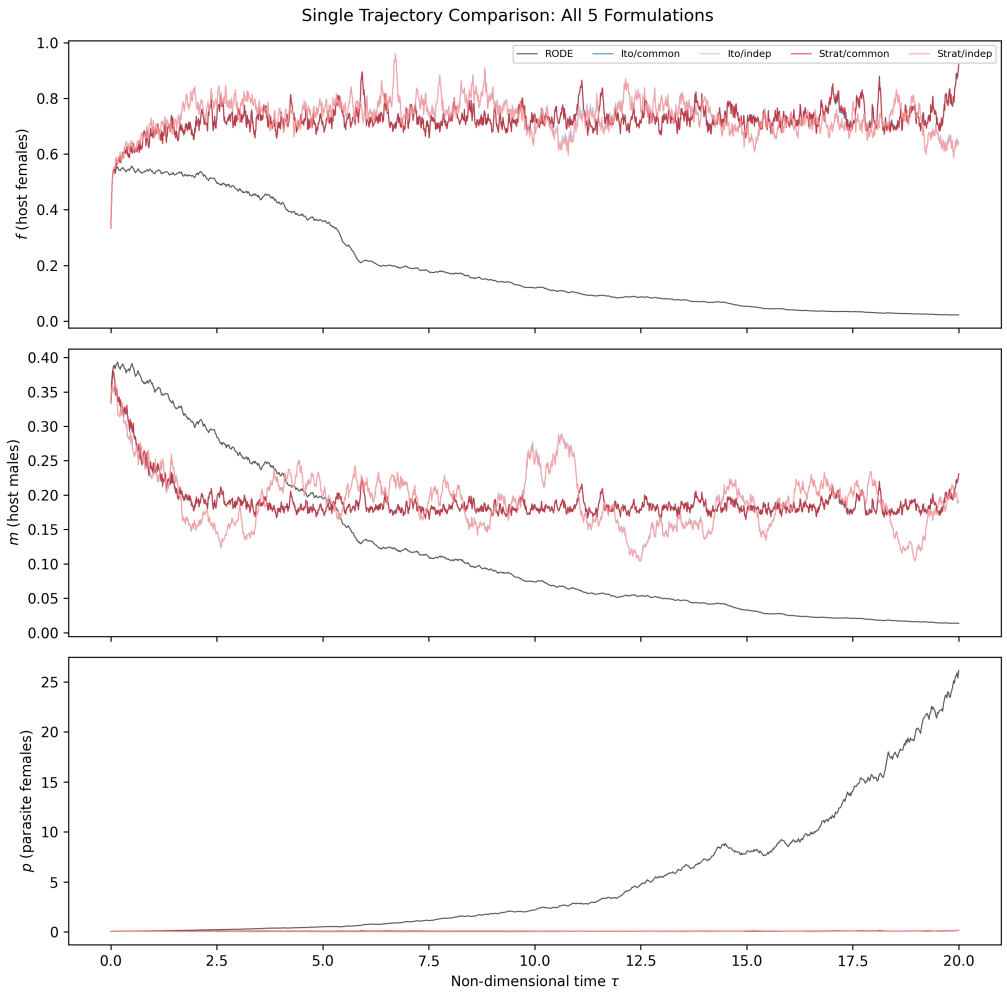
Scalar Lyapunov exponent:

$$\lambda_{\text{Strat}} = \beta_{\text{eff}} - \delta \quad \textbf{(independent of } \sigma_p\textbf{!)}$$

The Stratonovich correction $+\frac{1}{2}\sigma_p^2$ in the drift is exactly cancelled by the Itô correction $-\frac{1}{2}\sigma_p^2$ in $d \ln p$.

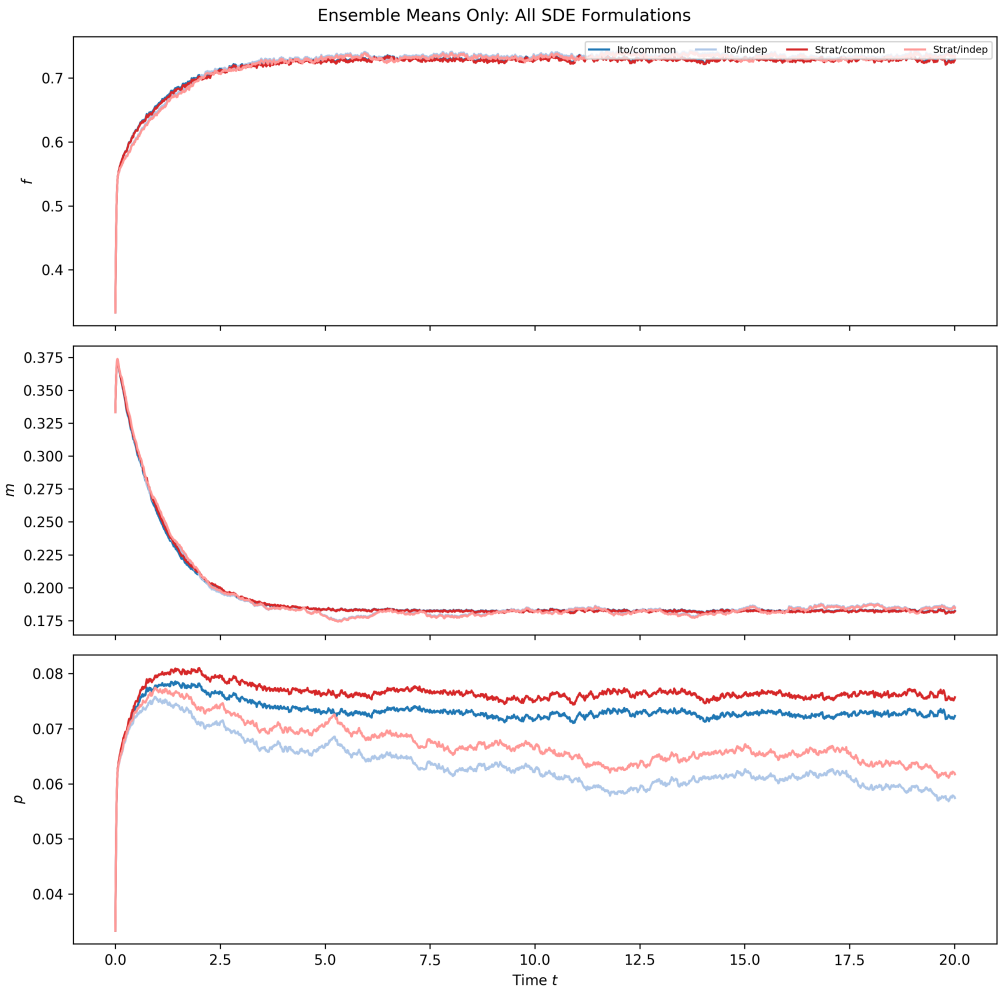
No amount of Stratonovich noise stabilizes the system by suppressing parasite growth.

Single Trajectory: All Five Formulations



Slow discrimination ($v = 6$): extinction in all formulations. Fast discrimination ($v = 60$): coexistence in all formulations. The *qualitative* outcome is robust; quantitative trajectories differ.

Ensemble Statistics: Mean $\pm$ 95% CI, $n = 500$ Paths



Stratonovich (red) sits systematically above Itô (blue) — the noise-induced mean shift predicted by Proposition 2. RODE (dark) differs in mean level due to positive mean noise bias ($\mathbb{E}[\eta] = \tilde{\eta}_{\max}/2 > 0$).

Moment Closure: What Is It and Why Do We Need It?

The moment hierarchy problem

Take expectations of the SDE for f:

$$\frac{d\mathbb{E}[f]}{dt} = \mathbb{E}\left[a\beta(1 - f - m - p)\, m\,(f - \gamma p)\right]$$

The right side contains $\mathbb{E}[fm]$, $\mathbb{E}[mp]$, $\mathbb{E}[fmp]$, ...

To find $\mathbb{E}[fm]$ we need $\mathbb{E}[f^2 m]$.
To find $\mathbb{E}[f^2 m]$ we need $\mathbb{E}[f^3 m]$.

The hierarchy never closes.

An exact ODE for means requires knowing all higher moments.

Mean-field closure: the approximation

Assumption: populations fluctuate nearly independently near the mean, so:

$$\mathbb{E}[f \cdot m] \approx \mathbb{E}[f] \cdot \mathbb{E}[m] = F \cdot M$$

Replace all products inside expectations with products of means.

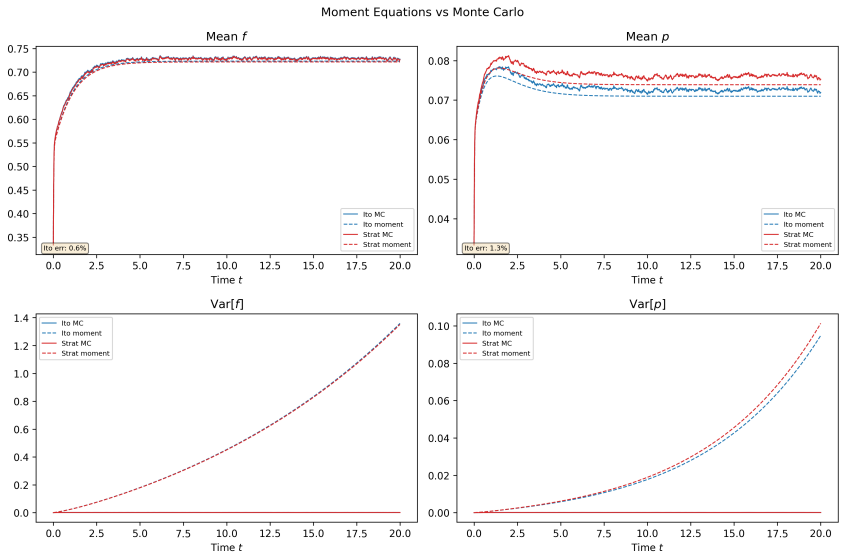
Result: the infinite hierarchy collapses to a *finite* ODE system in (F, M, P).

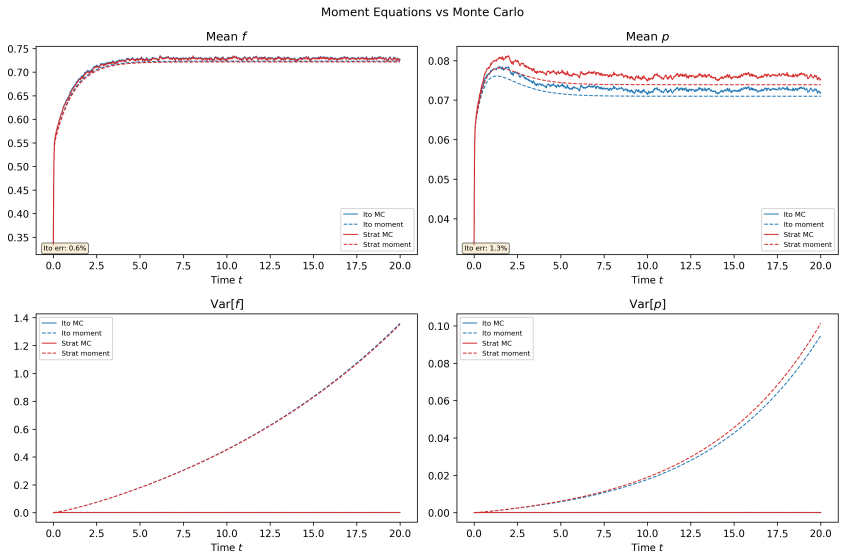
Exact when: f and m are uncorrelated.
Approximate when: correlations are weak relative to the mean trajectory.

What it gives us

- Analytical mean trajectories without Monte Carlo
- Explicit formula for noise-induced mean shift (Stratonovich)
- Validated against $n = 500$ path ensembles (previous slide)

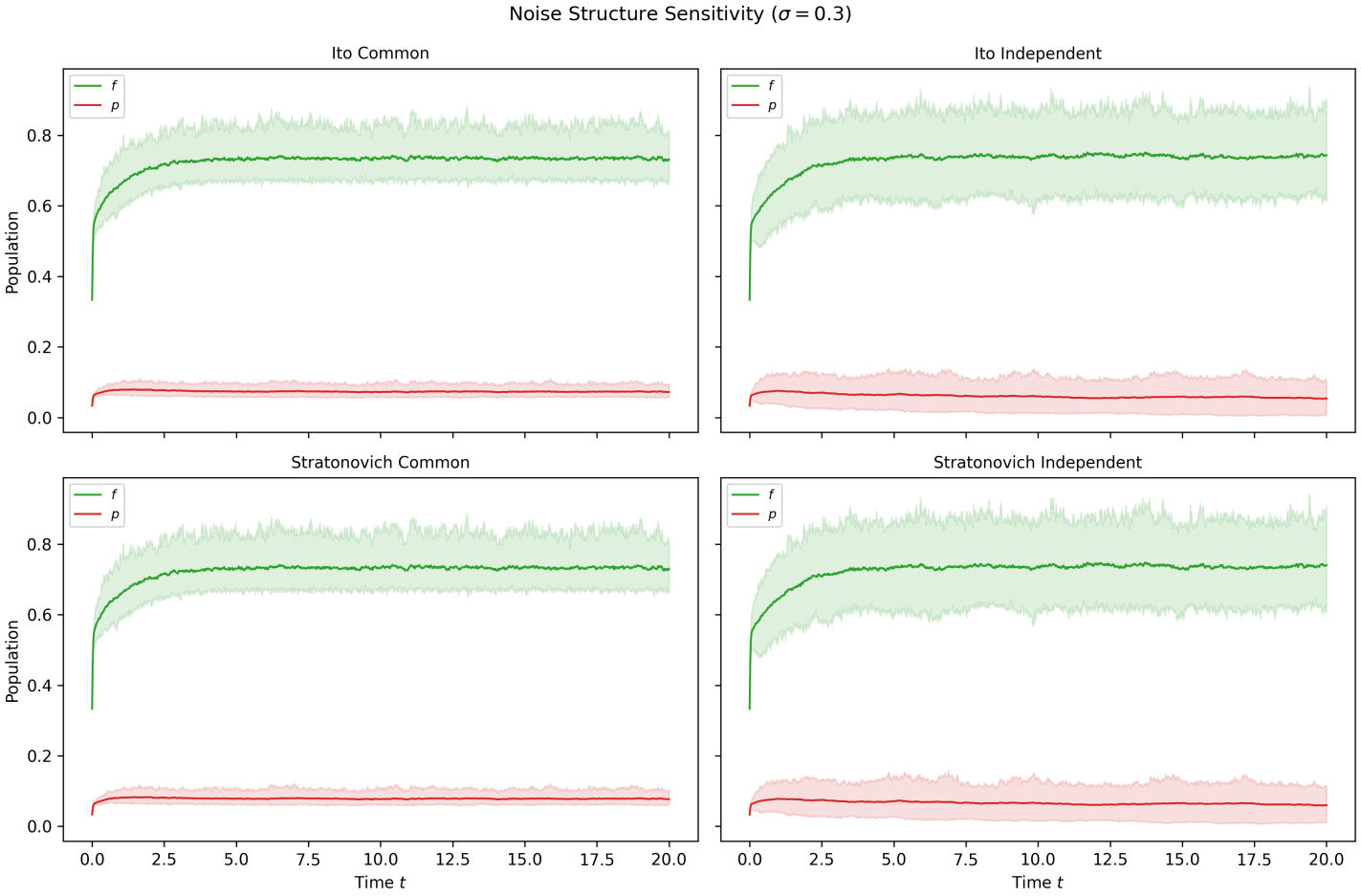
Moment Closure Validation





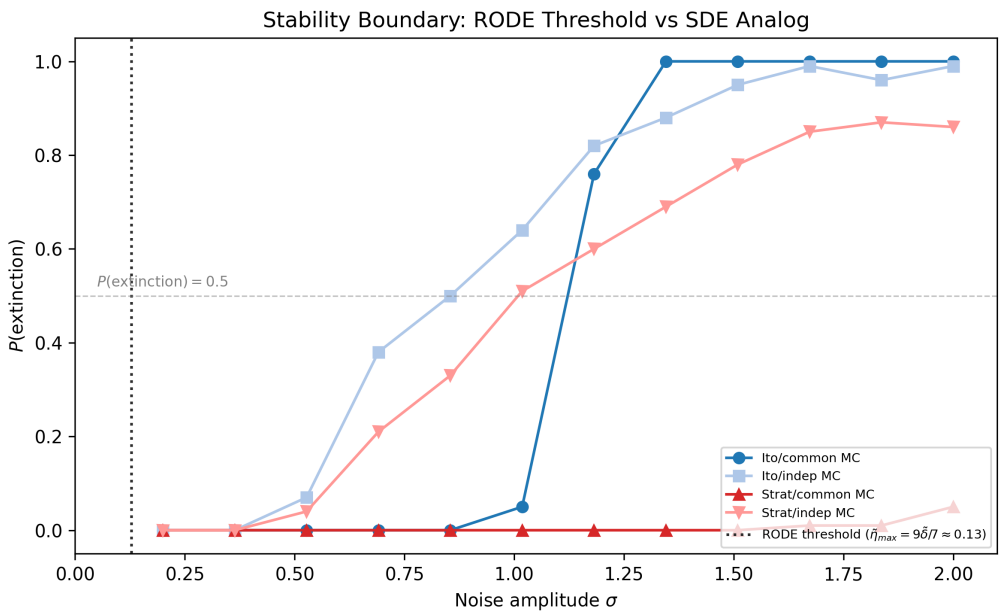
Dashed lines: moment equations under mean-field closure. Solid lines: Monte Carlo ensemble means ($n = 500$). **Left:** Itô/common noise — moment closure tracks MC well. **Right:** Stratonovich/common noise — upward shift confirmed analytically and numerically.

Noise Structure Sensitivity: Common vs. Independent



Common noise: covariance driven by $\sigma_f \sigma_m (FM + \mathrm{Cov}(f, m))$ — populations fluctuate coherently; cross-driving can reduce effective competition when $p > f$.
Independent noise: cross-driving term vanishes at mean-field level; deterministic coupling still generates covariance through shared nonlinear drift.

Stability Boundaries: All Five Formulations



Formulation	Noise	Boundary σ
RODE	—	$\tilde{\eta}_{\max} \approx 0.13$
Itô	Common	$\sigma \approx 1.12$
Itô	Indep.	$\sigma \approx 0.86$
Strat.	Common	> 2.0 (not reached)
Strat.	Indep.	$\sigma \approx 1.01$

RODE and SDE thresholds are analytically non-equivalent.

They measure fundamentally different things:

RODE: phase-space contraction at the population ceiling (Liouville / boundary dissipativity)

SDE: linearized growth rate of $\ln p$ near extinction (scalar Lyapunov exponent)

Direct comparison is ill-posed. Translating a RODE threshold into an SDE amplitude is mathematically undefined.

Single-Formulation Reporting: A Scientific Honesty Problem

A pattern in the stochastic modeling literature

A researcher studies a population system with stochastic environmental forcing. The biological noise mechanism is not fully characterized from field data. The researcher chooses one stochastic formulation — often Itô, because numerical libraries default to it — runs simulations, and reports extinction probabilities or stability boundaries as findings.

No justification is given for the formulation choice.
No competing formulations are tested.
The reported numbers are presented as evidence.

This is **not** a minor technical oversight. It is a form of **selective reporting**: the researcher is presenting one answer from a family of plausible models without disclosing that the family exists.

In this system: the Itô stability boundary is $\sigma \approx 1.12$; the Stratonovich/common boundary is $\sigma > 2.0$. These are **qualitatively different conclusions** about when noise stabilizes the system.

Robustness vs. Sensitivity: What Is Formulation-Dependent?

Formulation-**robust** results

- Discrimination speed v governs extinction vs. coexistence in all five formulations
- Slow $v \Rightarrow$ extinction; fast $v \Rightarrow$ coexistence *qualitatively* across all cases
- This ecological conclusion is not an artifact of formulation choice

Formulation-**sensitive** results

- Stability boundary σ differs significantly across formulations
- Extinction probability estimates diverge
- Mean trajectories differ (Itô $\neq$ Stratonovich)
- RODE dissipativity threshold has no SDE analog
- Noise-induced mean shift: Stratonovich only

What Data Would Discriminate Between Formulations?

Formulation choice	Discriminating data needed
Itô vs. Stratonovich *If $\tau_c \to 0$: Stratonovich (Wong-Zakai). If τ_c large: RODE more appropriate.*	Correlation time τ_c of environmental noise. *Measurable from hydrological records.*
Common vs. independent noise *If $\mathrm{Corr}(\delta f, \delta p) \approx 0$: independent.*	Cross-correlation of population fluctuations. *Measurable from multi-year census data.*
Displacement coefficient c *Riesch (2008) partial data: $c < 1$ likely. Requires controlled lab + field experiment.*	Sperm volume transferred per encounter type.

The modeling framework makes testable predictions.
The choice of formulation is not arbitrary — it is a hypothesis.

Summary

(a) **Discrimination speed v governs extinction/coexistence** across all five formulations — a robust ecological finding independent of stochastic framework choice.

(b) **RODE boundary dissipativity** $(\eta_f + \eta_m + \eta_p < 3)$ characterizes phase-space contraction at the carrying capacity. The Stratonovich Lyapunov exponent is noise-amplitude independent; the Itô exponent is shifted by $-\sigma^2/2$.

(c) **Formulation determines mean behavior:** Itô noise does not shift ensemble means (martingale property); Stratonovich noise adds an effective birth rate increment $+\frac{1}{2}\sigma_i^2\,\mathbb{E}[u_i]$ — largest for the parasite.

(d) **Under independent multiplicative noise,** diffusion cross-driving between populations vanishes at mean-field level; deterministic coupling still generates covariance through shared nonlinear drift.

Take-Home Message

On this biological system

Noise can save the sperm parasite — but *which* noise, formulated *how*, determines whether it does so by shifting the mean trajectory (Stratonovich) or by suppressing growth variance (Itô). These are different mechanisms with different empirical signatures.

On mathematical modeling practice

When the biological phenomenon is not fully characterized, choosing a single stochastic formulation and presenting its quantitative predictions as evidence is **scientifically dishonest**.

The minimum standard is: state your formulation, justify it biologically, and test the sensitivity of your conclusions to competing formulations.

Robustness to formulation choice is a result, not an assumption.

Thank you for your attention

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